

# A comparative acoustic study of English and Spanish vowels

Ann R. Bradlow<sup>a)</sup>

*Cornell University, Department of Modern Languages and Linguistics, Ithaca, New York 14853*

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Languages differ widely in the size of their vowel inventories; however, cross-linguistic surveys indicate that certain vowels and vowel system configurations are preferred. A cross-linguistic comparison of the acoustic vowel categories of two languages that differ in vowel inventory size, namely, English and Spanish, was performed in order to reveal some of the language-specific and/or universal principles that determine the acoustic realization of the vowels of these two languages. This comparison shows that the precise location in the acoustic space of similar vowel categories across the two languages is determined, in part, by a language-specific base-of-articulation property. These data also suggest that the relatively crowded acoustic vowel space of English may be expanded with respect to the relatively uncrowded acoustic vowel space of Spanish; however, this effect is variable depending on the syllable context of the English vowels. Finally, the data indicate no difference in the tightness of within-category clustering for the large versus the small vowel inventory.

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## INTRODUCTION

Surveys of segment inventories indicate cross-linguistic preferences for certain vowels and for certain vowel inventory configurations. For example, in a survey of 317 languages, Maddieson (1984) finds that vowel inventories in this sample vary from having three to fifteen distinct vowel qualities, with two-thirds of the languages having between five and seven distinct vowel qualities. Additionally, the specific vowels that comprise these statistically preferred vowel inventories tend to be the same. For example, five-vowel systems tend to have /i,e,a,o,u/, seven-vowel systems tend to have these five vowels plus /ɛ/ and /ɔ/, and six-vowel systems usually have /i,e,æ,ɑ,o,u/. Furthermore, the vowel inventories of the vast majority of the world's languages include the three vowels that define the extremes of the general vowel space, namely /i,a,u/. Accordingly, these three vowels are known as the "point vowels," and have been afforded a special status in theories of vowel systems.

These cross-linguistic tendencies have led to the hypothesis that there are constraints on possible speech sounds and their cooccurrence, which have their source in general linguistic, or physical (i.e., auditory and articulatory) constraints. However, the exact nature of these constraints and their interaction that produces the observed inventories is not yet fully understood. This study is thus motivated by a general interest in the effect of inventory size on the acoustic vowel spaces of different languages, and in how this effect might reveal some of the universal and/or language-specific constraints leading to the observed patterns in sound inventories of the languages of the world. Specifically, this paper explores the effect of inventory size on the acoustic realization of vowels in a language with a relatively large vowel inventory, General American English, and in a language with a relatively small vowel inventory, Madrid Spanish.

## I. GENERAL APPROACHES TO THE STUDY OF VOWEL INVENTORIES

Previous work has led to the development of several theoretical positions regarding the structure of vowel systems. Dispersion theory (DT) claims that speech sounds are selected via constraints that are based on a principle of sufficient perceptual contrast. In this theory the vowels of a given language are arranged in the acoustic vowel space so as to minimize the potential for perceptual confusion between the distinct vowel categories. Using computer programs to generate the optimal configurations for vowel systems of various sizes, this approach to vowel inventories has proved fairly successful (Liljencrants and Lindblom, 1972; Lindblom, 1975, 1986; Disner, 1984). However these investigations of DT focus exclusively on intercategory distance as the determiner of vowel system configuration in a universally defined acoustic vowel space. As a result, this approach fails to account for the observation that certain languages, such as Swedish with nine vowels and Danish with ten vowels, crowd their vowels into a small corner of the entire vowel space rather than dispersing them throughout the available space (Disner, 1983). In more recent developments, the dispersion principle has been expressed as a principle of sufficient, rather than maximal, contrast (Lindblom, 1989, 1990). Furthermore the theory has been extended to account for within-speaker variation. For example, Moon and Lindblom, (1989) show that under circumstances that require clear speech, a speaker's vowel space will be expanded relative to his or her casual speech vowel space.

In a study that addresses the prediction of DT that vowels will be maximally dispersed in the acoustic space, Jongman *et al.* (1989) compared the relatively crowded vowel spaces of English (with 11 monophthongs) and German (with 14 monophthongs) with the relatively uncrowded vowel space of Greek (with just 5 monophthongs). These authors found that the crowded vowel spaces of English and German are expanded relative to the uncrowded vowel space

<sup>a)</sup>Current affiliation: Speech Research Laboratory, Psychology Department, Indiana University, Bloomington, IN 47405.

of Greek.<sup>1</sup> Jongman *et al.* plotted the vowels of these languages in the auditory-perceptual space proposed by Miller (1989). This representation scheme is designed to normalize the data points for both inter- and intraspeaker differences by representing speech sounds in terms of log ratios of the fundamental frequency and the first three formants. The results of this study support the hypothesis that the acoustic realization of vowel categories is dependent on inventory size and suggest that some version of the dispersion principle does indeed hold true across languages: the larger the inventory, the more "expanded" the acoustic vowel space.

The quantal theory of speech (QTS) (Stevens, 1972, 1989) suggests an alternative approach to vowel systems. This theory is based on the observation that for certain parameters of the articulatory domain, there is a nonmonotonic relation between variation in the articulatory configuration and its acoustic consequences. Similarly, certain changes in the acoustic signal over some part of the range of a particular parameter are nonmonotonically related to the corresponding auditory response of the listener. In other words, according to this theory, there are certain regions of stability in the phonetic space. In particular, it is claimed that there are stable regions corresponding to the point vowels /i/, /a/, and /u/. Thus this theory predicts that the point vowels should be in approximately the same locations across all languages, regardless of vowel inventory size. Furthermore, QTS predicts that, since the point vowels are in phonetically stable regions, they should show less within-category variability than nonpoint vowels.

Evidence that certain changes in the articulatory domain are nonmonotonically related to their acoustic consequences comes from the observation that there are acoustic properties which are relatively insensitive to articulatory perturbation. For example, in the case of nonlow front vowels which have a fronted tongue body, the locations in the frequency spectrum of the second, third, and fourth prominences are relatively insensitive to perturbation in the tongue-body position along the anterior-posterior dimension (Stevens, 1989). For this range,  $F_2$  is at a maximum and is within a few hundred hertz of  $F_3$ . In contrast, the frequency of  $F_1$  varies monotonically with the size and position of the articulatory constriction. This result is presented by Stevens as evidence of a stable region in this corner of the vowel space, which corresponds to the point vowel /i/. Furthermore, in an articulatory study of /i,a,u/ in General American English, Perkell and Cohen (1989) find that for each of these vowels there is both an articulation-to-acoustic saturation effect, and a muscle contraction-to-displacement saturation effect. In other words, these authors find that over a range of changes in the articulation of these vowels, the acoustic output is relatively stable.

DT and QTS both propose general universal principles to account for the observed cross-linguistic tendencies regarding vowel inventory size and structure. In contrast to these approaches, the notion of a language-specific base-of-articulation is presented as an account for the observation that similar sounds across two languages can differ due to a consistent, language-specific adjustment of the articulators. This notion has been a part of the traditional phonetic literature over the ages: Disner (1983) cites its origin as the work

of John Wallis in 1653 (see Kemp, 1972). However, within the tradition of generative phonology, the idea of a language-specific articulatory setting has often been considered outside of the area of interest of theoretical linguistics. For example, in *The Sound Pattern of English*, Chomsky and Halle (1968) consider this aspect of speech as extragrammatical, and thus as part of the performance aspect of language, rather than part of the grammatically determined competence aspect.

Nevertheless, several investigators have pointed to the importance of the notion of a base-of-articulation for providing insightful analyses of both phonological and phonetic observations. In studies that have tested the predictions of DT and QTS, phonetic differences between similar segments of different languages have often been observed. For example, Lindau and Wood (1977) investigate the vowels of three related Nigerian languages, Yoruba, Edo, and Ghotuo, all of which have phonemically equivalent seven-vowel systems, and find that the vowel spaces of Edo and Ghotuo are very similar. However, contrary to the prediction of the dispersion principle, the vowel space of Yoruba deviates from the structure of the other two seven-vowel systems and is not maximally dispersed. Similarly, Disner (1983) finds that the seven-vowel systems of Yoruba and Italian differ from each other in their locations of the seven vowels in the acoustic space. Disner documents additional cases of systematic differences across the vowels of several Germanic languages; for example, she finds that the vowels of Danish are systematically articulated with a higher tongue position, as it is reflected by  $F_1$ , than the vowels of English. Disner claims that these data demonstrate the role of a language-specific base-of-articulation property in the phonetic realization of vowel phonemes. In particular, this type of language-specific effect is seen in across-the-board shifts of the vowels of one language relative to similar vowels in another language.

In light of the theoretical claims and experimental evidence discussed above, the present study was undertaken as a direct means of assessing the contributions of language-specific and general universal principles in the acoustic realization of vowel categories across languages with relatively large versus small vowel inventories. English and Spanish were chosen for this study because of the large difference between the sizes of their vowel inventories: English has more than double the number of stressed monophthongal vowels than Spanish. Additionally, the five-vowel system of Spanish is statistically very common, whereas the 11-vowel system of English is uncommonly large (Maddieson, 1984). Furthermore, the vowel systems of these two languages are similar in that they vary along the same dimensions (neither language has contrastive rounding, length, or nasalization). Consequently, the principal difference between these vowel systems is in the number of vowels. Thus this English-Spanish comparison represents a comparison of an unusually large vowel inventory with a smaller cross-linguistically common vowel inventory. The expectation is that this difference between the two vowel inventories will highlight both the differences and similarities attributable to language-specific and/or universal aspects of vowel production.

In this comparison, vowel formant measurements from each of the two languages are evaluated in terms of the pre-

TABLE I. Spanish CVCV, English CVC, and English CVCV mean vowel formants in hertz with standard deviations.

|   | Spanish CVCV |           |   | English CVC |           |   | English CVCV |           |
|---|--------------|-----------|---|-------------|-----------|---|--------------|-----------|
|   | F1 (s.d.)    | F2 (s.d.) |   | F1 (s.d.)   | F2 (s.d.) |   | F1 (s.d.)    | F2 (s.d.) |
| i | 286(6)       | 2147(131) | i | 268(20)     | 2393(239) | i | 264(34)      | 2268(207) |
| e | 458(42)      | 1814(131) | ɪ | 463(34)     | 1995(199) | ɪ | 429(20)      | 1831(173) |
| a | 638(36)      | 1353(84)  | e | 430(45)     | 2200(168) | e | 424(39)      | 2020(149) |
| o | 460(19)      | 1019(99)  | ɛ | 635(53)     | 1796(149) | ɛ | 615(60)      | 1665(143) |
| u | 322(20)      | 992(121)  | æ | 777(81)     | 1738(177) | æ | 773(62)      | 1640(169) |
|   |              |           | ʌ | 640(39)     | 1354(134) | ʌ | 655(43)      | 1216(75)  |
|   |              |           | ɑ | 780(83)     | 1244(145) | ɑ | 783(155)     | 1182(152) |
|   |              |           | ɔ | 620(72)     | 1033(135) | ɔ | 614(65)      | 945(83)   |
|   |              |           | o | 482(30)     | 1160(47)  | o | 473(13)      | 1094(41)  |
|   |              |           | ʊ | 481(36)     | 1331(161) | ʊ | 411(17)      | 1361(94)  |
|   |              |           | u | 326(26)     | 1238(160) | u | 316(43)      | 1183(153) |

dictions of the various approaches to vowel inventories. First, the locations of the four common vowels are compared in order to reveal any base-of-articulation effect. Second, the range and degree of within-category variance for vowels in each of the languages are compared in order to determine the effect of inventory size on the general dispersion of the vowel categories. In general, these acoustic data are analyzed in relation to the proposals in the literature, and with regard to identifying the language-specific and language-independent factors that affect the acoustic realization of vowels.

## II. METHOD

Words exemplifying the 11 vowel contrasts of English and the five vowel contrasts of Spanish were selected such that the target vowels all occur between either /p/ or /b/, and /t/. The English words are all monosyllabic (*beat, bit, bait, bet, bat, pot, bought, boat, put, boot, but*), and in accordance with Spanish phonotactics and syllabification, the Spanish words are disyllabic (*bita, beta, bata, bota, puta*). The words were embedded in frame sentences that are similar in length, syntactic structure, and position of the target word across the two languages. The English and Spanish frame sentences were "Say \_ again" and "Escribe \_ bien" ("Write \_ well"), respectively. A list of these sentences was constructed for each language such that each sentence was repeated five times in random order, and the subjects were instructed to read the sentence list from their respective languages with normal speed and intonation. Four male speakers of General American English and four male speakers of Madrid Spanish served as subjects, giving a total of 20 tokens (four speakers  $\times$  five repetitions) for each vowel category of each language. All of the English speakers have spent most of their lives in the Ithaca, New York area. All of the Spanish speakers come from Madrid: two of these have spent significant amounts of time in the U.S.; the other two have spent almost all of their lives in the Madrid area.

All recordings were made with a portable cassette recorder (Marantz PMD222) and an AKG D310 microphone. The English recordings were made in a sound-attenuated booth in the Phonetics Laboratory at Cornell University. The Spanish recordings were made in a quiet room in Madrid. The recordings were digitized with a sampling rate of 12 000

Hz and low-pass filtered at 6000 Hz. All measurements were made using the Entropics WAVES+ speech analysis software on a SUN workstation. Both LPC spectra and spectrograms were used to determine the first three formant frequencies. The LPC spectra were calculated from a 25-ms Hanning window in the vowel steady state. In most cases an analysis order of 14 was used; however, in a small number of cases the analysis order was lowered to 12 so that the data were smoothed to yield a clearer peak. The steady-state portion of the vowel was determined by first placing two vertical cursors in the spectrogram: the first cursor marked the end of the formant transitions coming out of the initial consonant, and the second cursor marked the beginning of the formant transitions into the final consonant. The period between the two cursors thus indicated the portion of the vowel which showed no (or very little) formant movement. A 25-ms Hanning window was placed approximately in the middle of this steady-state period. The formant values were then read from the LPC spectrum and checked with readings from the spectrogram. In the cases of English /e/ and /o/, which are sometimes diphthongized, the formant measurements were taken from the portion of the vowel before the offglide. The auditory quality of this portion of the vowel was assessed, in order to insure that the formant measurements being recorded were those appropriate for /e/ or /o/ rather than for the /j/ or /w/ offglides. The fundamental frequency was found by taking the inverse of the mean of three peak-to-peak durations from the central portion of the target vowel waveform.

## III. RESULTS

Table I gives the mean *F1* and *F2* values with standard deviations for the data from the four male Spanish speakers (Spanish CVCV) and from the four male English speakers (English CVC).<sup>2</sup> In this table, and in all subsequent analyses of these data, the standard deviations represent the variance of the means of the five tokens for each of the four speakers for each language. The measurements of the five tokens within each subject are treated as repeated measures and therefore averaged for the purposes of the cross-language comparisons.

The following subsections discuss these results and their bearing on the approaches to vowel inventories discussed in Sec. I above.

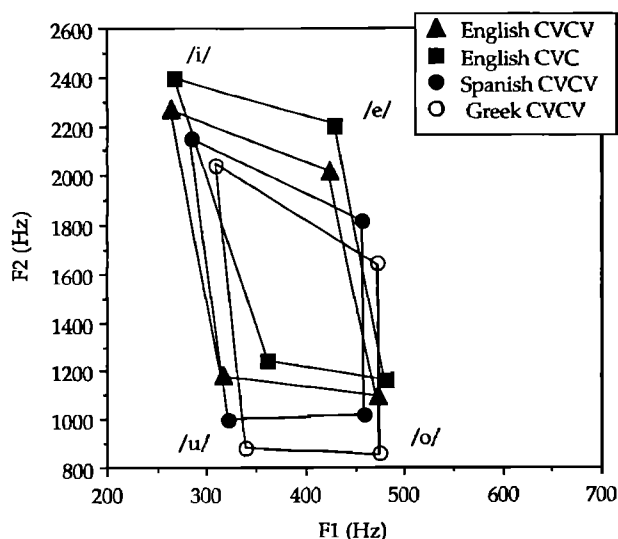


FIG. 1. Comparison of the areas in the  $F1 \times F2$  space covered by English (CVC and CVCV contexts), Spanish, and Greek /i/, /e/, /o/, and /u/. Greek data from Jongman *et al.* (1989).

## A. Base of articulation

The first issue addressed by this acoustic study concerns the general placement of the English and Spanish vowels in the acoustic vowel space. In this regard we might expect both a language-specific effect, which will cause similar vowel categories across two languages to differ in a systematic way due to a consistent language-specific adjustment of the articulators, as well as a general expansion effect for languages with crowded inventories relative to languages with uncrowded inventories. These two possible effects on vowel location would necessarily interact with each other; thus it is important to examine each individually in order to assess the extent of the two separate effects. In this section, then, the focus is on assessing the presence (and extent) of any language-specific base-of-articulation effect, which may result in a general shift in some direction of all the vowels of one language relative to the equivalent vowels of the other language.

A comparison of the locations of the common vowels across English and Spanish, namely /i/, /e/, /o/, and /u/, indicates a general upward shift in the  $F2$  dimension of the English vowels relative to the Spanish vowels (see Fig. 1). The low vowel is excluded from the set of common vowels because for these Spanish speakers, the low vowel is central in the front-back dimension (as represented by the IPA symbol /a/), whereas, for these English speakers, there is a low front vowel (IPA /æ/) and a low back vowel (IPA /ɑ/) but no low central vowel. A two-factor ANOVA with language (Spanish, English) and vowel (the four common vowels) as factors, and  $F2$  as the dependent variable, confirms this impression by showing a significant main effect for language [ $F(1,3)=24.17$ ,  $p<0.001$ ]. The language  $\times$  vowel interaction is not significant [ $F(1,3)=0.942$ ,  $p=0.4358$ ].

The effect of this difference in  $F2$  between the English and Spanish vowels is that the English front vowels, namely /i/ and /e/, are located more peripherally in the acoustic space than the Spanish front vowels. This effect on the front vowels

would thus seem to concur with the result we might expect from an expansion effect. However, we also find that the English back vowels, namely /u/ and /o/, are raised in the  $F2$  dimension relative to the Spanish back vowels, resulting in a more central location for the English back vowels. This result for the back vowels is in contradiction to the result we might expect from an expansion effect, since such an effect should result in English back vowels with lower  $F2$  frequencies than their Spanish counterparts. This consistent upward shift in the  $F2$  dimension of the English vowels relative to the Spanish vowels is in accordance with the result we would expect from a language-specific base-of-articulation effect. Specifically, this result suggests that the vowels of English are articulated with a fronted tongue position relative to the tongue position for the Spanish vowels.

An alternative explanation for the higher  $F2$  of the English vowels relative to the Spanish vowels is that the four individual English speakers in this study all have shorter vocal tracts than the four individual Spanish speakers, resulting in generally higher formants for the English speakers than for the Spanish speakers. In such a situation, the present data would also show a systematic raising of the  $F1$  frequencies for the English speakers relative to the  $F1$  frequencies for the Spanish speakers. However, if the present data do not show a similar effect for  $F1$  as for  $F2$ , then we can rule out the possibility that the difference in  $F2$  for the two groups of speakers is a function of vocal tract length.

As seen in Fig. 1, English /i/ is lower in  $F1$  than Spanish /i/, and English /e/ tends towards a lower  $F1$  than Spanish /e/. However, for the back vowels, English /o/ is higher in  $F1$  than Spanish /o/, and English /u/ and Spanish /u/ do not show any difference in  $F1$ . Indeed, in an ANOVA comparing the  $F1$  frequencies in English and Spanish for the four common vowels, the main effect of language [ $F(1,3)=0.277$ ,  $p=0.603$ ] fails to reach significance, as does the language  $\times$  vowel interaction [ $F(1,3)=1.236$ ,  $p=0.3184$ ]. Thus in the  $F1$  dimension we do not find a consistent effect as we do in the  $F2$  dimension. Since these data do not indicate a systematic raising of all the English formants relative to the Spanish formants, an account of different vocal tract lengths between the two groups of speakers can be ruled out.

## B. English CVCV tokens

Recall that the English vowel tokens all occurred in monosyllabic words whereas the Spanish tokens all occurred in disyllabic words. It is thus possible that the consistent shift in  $F2$  of the English vowels relative to the Spanish vowels is due to a more general effect, such as the syllabic structure of the test words, rather than due to a language-specific base-of-articulation property. For instance, it is possible that the tautosyllabic coronal, with its characteristically high  $F2$  locus, in the English test words raised the  $F2$  of the target vowel, especially in the case of the back vowels which are generally characterized by low second formants. Therefore in order to assess the effect of word structure, a new set of English recordings was made using disyllabic nonwords that matched the syllabic structure, as well as the segmental structure, of the Spanish test words. This new set of data thus

allowed for a direct comparison of the tokens with the same syllabic structure as produced by speakers of the two languages.

Three of the four male speakers of General American English from the first experiment produced disyllabic non-word versions of the tokens used in the original data set. These test tokens were embedded in the same frame sentence as used in the first experiment, and the speakers read each sentence five times in random order. The speakers read these sentences from a list that was constructed as follows: the word from the original list that contained the target vowel appeared on one line, and the frame sentence with the target nonword appeared on the following line. In order to reinforce the disyllabicity of the test tokens, and to avoid the production of a flap for the medial /t/, the target nonword was typed with a period separating the two syllables (e.g., *bea.ta*), and the speakers were instructed that the period indicated a syllable boundary and to avoid producing a flap for the intervocalic /t/. This procedure was effective in eliciting tokens that matched their Spanish counterparts at both the segmental and intonational levels.

This set of data was collected and analyzed identically to the original set of English data. The mean  $F1$  and  $F2$  values and standard deviations for these English disyllabic tokens are given in Table I (English CVCV). These English CVCV tokens were then compared to the Spanish CVCV tokens to see if the systematic, cross-language  $F2$  difference that was obtained in the previous experiment persists under conditions of similar word structure.<sup>3</sup> Indeed, ANOVAs comparing the four shared vowels of English and Spanish in CVCV contexts show that there is a significant main effect of language, such that English vowels are generally higher in  $F2$  than the Spanish vowels [ $F(1,3)=8.483$ ,  $p=0.0086$ ]. Furthermore, as in the previous experiment, this main effect fails to reach significance in the  $F1$  dimension [ $F(1,3)=1.232$ ,  $p=0.2803$ ]. Finally, the language  $\times$  vowel interaction is not significant for either  $F1$  [ $F(1,3)=0.819$ ,  $p=0.4984$ ] or  $F2$  [ $F(1,3)=0.361$ ,  $p=0.7816$ ]. Thus these data indicate that the effect we found in the comparison of the English monosyllabic words and the Spanish disyllabic words cannot be accounted for by the difference in word structure alone. Rather, these results suggest that indeed the English vowels differ systematically from the Spanish vowels, in a manner that is consistent with the notion of a language-specific base-of-articulation effect.

In summary, the crucial point in this comparison of the English and Spanish vowels is that all the vowels of English are generally higher in  $F2$  than their Spanish counterparts. This systematic difference for all of the shared vowels of the two languages indicates a language-specific base-of-articulation effect, rather than an expansion effect due to the crowdedness of the English vowel space relative to the Spanish vowel space. Furthermore, the observed  $F2$  difference cannot be accounted for by a difference in vocal tract length between the two groups of speakers, since we do not find similar shifts for all formants. Finally, an effect of word structure can be discounted since the observed formant differences across the two languages persist when word structure is controlled. Thus this comparison of vowel locations

across English and Spanish indicates the presence of a language-specific factor in the phonetic realization of phonemic vowel categories.

### C. A comparison of two five-vowel systems

In order to test the independence of this base-of-articulation effect from an effect of inventory size, a comparison of two phonemically equivalent five-vowel systems was performed. For this comparison, the present Spanish data were compared to data for Greek vowels taken from Jongman *et al.* (1989). In that study, the authors recorded four male native speakers of Modern Greek producing four repetitions of words exemplifying the five phonemic vowel contrasts of the language. In these bisyllabic words, the target vowels were preceded by a bilabial consonant and followed by an alveolar consonant, thus facilitating a direct comparison with the present Spanish data which used the same consonantal context. The words were embedded in frame sentences, and the vowel formant measurements were taken from LPC spectra, as described in that paper.

A visual comparison of the vowels in Spanish and Greek (see Fig. 1) shows that the vowels of Spanish are generally higher in the  $F2$  dimension than their Greek counterparts, and, in the  $F1$  dimension, there is a general trend for the Spanish vowels to have lower values than the Greek vowels. ANOVAs with language (Spanish, Greek) and vowel (/i/, /e/, /a/, /o/, /u/) as factors confirm these visual impressions: for both  $F1$  and  $F2$  there is a significant main effect of language [ $F(1,4)=5.565$ ,  $p=0.025$  for  $F1$ ,  $F(1,4)=9.913$ ,  $p=0.0037$  for  $F2$ ]. Additionally, in both cases the language  $\times$  vowel interaction fails to reach significance [ $F(1,4)=0.264$ ,  $p=0.899$  for  $F1$ ,  $F(1,4)=0.100$ ,  $p=0.9814$  for  $F2$ ].

The general result of this comparison of the Greek and Spanish vowels is that these two phonemically equivalent vowel systems show a systematic difference regarding the acoustic realization of the shared vowel categories: the vowels of Spanish are generally higher in  $F2$ , and lower in  $F1$  than the corresponding vowels of Greek. Thus we can conclude that a language-specific, base-of-articulation property plays an important role in determining the location of vowel categories in the acoustic space, and that this property functions independently of the general size and structure of the vowel inventory. In particular, the present data indicate that the phonemically equivalent vowels of English, Greek, and Spanish all differ systematically with respect to one another in the  $F2$  dimension. In the case of each of the four vowels common to Greek, Spanish, and English, the English vowel is higher in  $F2$  than the Spanish vowel, which is in turn higher in  $F2$  than the Greek vowel. These observations thus suggest that, with respect to these three languages, the general base-of-articulation for English involves the most fronted tongue position, the Spanish base-of-articulation involves an intermediate tongue position, and the Greek base-of-articulation involves the least fronted tongue position.

TABLE II. Comparison of the /i/-e/-o/-u/ area in English, Spanish, and Greek.

|                              | Difference              | Difference |
|------------------------------|-------------------------|------------|
| English CVC vs Spanish CVCV  | +18 832 Hz <sup>2</sup> | +12.7%     |
| English CVC vs Greek CVCV    | +23 749 Hz <sup>2</sup> | +16.6%     |
| English CVCV vs Spanish CVCV | +2645 Hz <sup>2</sup>   | +1.8%      |
| English CVCV vs Greek CVCV   | +7562 Hz <sup>2</sup>   | +5.3%      |
| Spanish CVCV vs Greek CVCV   | +4917 Hz <sup>2</sup>   | +3.4%      |
| English CVC vs English CVCV  | +16 187 Hz <sup>2</sup> | +10.7%     |

#### D. Expansion of the acoustic vowel space

I now turn to a comparison of the range, or area in the acoustic vowel space, covered by the vowel categories of English and Spanish. Based on the dispersion principle, we expect that the relative crowdedness of the English vowel inventory will cause an expansion of the English acoustic vowel space relative to the Spanish acoustic vowel space. In the comparison of the locations of the English and Spanish vowels in the acoustic space we saw that the English vowels are systematically shifted upward in the *F2* dimension relative to the Spanish vowels. However, this difference between the two languages regarding their bases-of-articulation does not preclude an expansion effect. In other words, it is possible that the English vowels are both higher in *F2* and cover a greater area than the Spanish vowels.

In order to compare the general range of the English and Spanish vowels, the area covered by the quadrilaterals defined by the mean *F1* and *F2* values of the four common vowels was calculated (Table II). For the English vowels, both the CVCV and the CVC data were included in this comparison. As an additional point of comparison, the corresponding area covered by the Greek /i/, /e/, /o/, and /u/ was included in this analysis. Figure 1 shows these quadrilaterals in the *F1* by *F2* spaces of each of the three languages.

The results of this comparison of the /i/-e/-o/-u/ area in English, Spanish, and Greek show that the English acoustic vowel spaces, for both the CVCV and CVC data, cover more area than either the Spanish or the Greek acoustic vowel spaces. However, the magnitude of the difference varies depending on the syllabic context of the English vowels. The area covered by English /i/-e/-o/-u/ in the closed syllable context (the CVC data) is 12.7% greater than the Spanish area, and 16.6% greater than the Greek area, whereas the area covered by English /i/-e/-o/-u/ in the open syllable context (the CVCV data) is only 1.8% greater than the Spanish area, and 5.3% greater than the Greek area. (The /i/-e/-o/-u/ areas in Spanish and Greek differ by 3.4%.) Thus the English closed syllable vowel space is expanded in the acoustic domain relative to the Spanish and

Greek vowel spaces, whereas, in the open syllable context, the three vowel spaces cover comparable areas.

#### E. Tightness of within-category clustering

In order to investigate the effect of number of phonological categories on tightness of within-category clustering, the coefficients of variations for the *F1* and *F2* values of the four vowels common to English and Spanish were compared. For this comparison the English CVC (closed syllable) data were compared to the Spanish CVCV data. In addition, the English CVC (closed syllable) data were compared to the English CVCV (open syllable) data. Both of these comparisons investigate the effect of many versus few categories: the former is within English and across syllabic contexts; the latter is across languages using syllabic contexts that contrast many versus few categories. Table III gives the results of this comparison.

The present data also provide a means for testing theoretical predictions regarding differences in clustering between categories within each language. Specifically, according to QTS (Stevens, 1972, 1989) the point vowels /i,a,u/ should show less within-category variability since they occur in quantal regions of the articulatory vowel space. However, in a test of this prediction, Pisoni (1980) found that the standard deviations of the English point vowels were not significantly smaller than the standard deviations of the nonpoint vowels. In accordance with that result, based on the present data, we find that the *F1* and *F2* standard deviations of English /i,a,u/ are not significantly smaller than the *F1* and *F2* standard deviations of the eight English nonpoint vowels. (For this comparison, the English CVC and CVCV data were pooled.<sup>4</sup>) Similarly, we find no difference between the *F1* and *F2* standard deviations for the Spanish point and nonpoint vowels. The results of these analyses are summarized in Table IV.

The results of these comparisons indicate that the tightness of within-category clustering does not vary significantly as a function of the number of phonological categories. The *F1* and *F2* coefficients of variation do not differ significantly across the English CVC data and the Spanish CVCV data, nor do they differ within English across syllabic contexts. Furthermore, we find no difference between the point vowels and nonpoint vowels with regard to within-category variation. Thus these data indicate that the tightness of within-category clustering is not dependent on the size of the vowel inventory.

TABLE III. A comparison of the *F1* and *F2* coefficients of variation across English CVC and Spanish CVCV (/i,e,o,u/), and English CVC and English CVCV (all 11 phonetic categories).

|           | English CVC vs Spanish CVCV |       |        |              |                | English CVC vs English CVCV |       |        |               |                |
|-----------|-----------------------------|-------|--------|--------------|----------------|-----------------------------|-------|--------|---------------|----------------|
|           | Eng.                        | Span. | diff.  | <i>t</i> (3) | <i>p</i> value | CVC                         | CVCV  | diff.  | <i>t</i> (10) | <i>p</i> value |
| <i>F1</i> | 0.094                       | 0.065 | 0.029  | 2.165        | 0.0735         | 0.086                       | 0.092 | -0.006 | -0.48         | 0.6413         |
| <i>F2</i> | 0.089                       | 0.094 | -0.005 | -2.02        | 0.8468         | 0.092                       | 0.087 | 0.005  | 0.782         | 0.4523         |

TABLE IV. A comparison of the *F1* and *F2* standard deviations for the point vs nonpoint vowels in English (CVC and CVCV data) and in Spanish.

|           | pt. | non-pt. | English<br>diff. | <i>t</i> (20) | <i>p</i> value | pt. | non-pt. | Spanish<br>diff. | <i>t</i> (3) | <i>p</i> value |
|-----------|-----|---------|------------------|---------------|----------------|-----|---------|------------------|--------------|----------------|
| <i>F1</i> | 60  | 44      | 16               | 1.072         | 0.2964         | 26  | 31      | 5                | 0.422        | 0.7014         |
| <i>F2</i> | 176 | 131     | 45               | 2.048         | 0.0539         | 126 | 111     | -15              | -0.522       | 0.6375         |

## IV. DISCUSSION

### A. Base of articulation

The general goal of this comparative acoustic study of the English and Spanish vowels was to explore the effect of inventory size on the acoustic realization of the phonemic vowel categories of English and Spanish. The first finding of this study is that the English and Spanish vowel spaces differ systematically in the location of their vowel categories in the acoustic space defined by *F1* and *F2*. A comparison of the locations of the four vowels common to English and Spanish, namely /i/, /e/, /o/, and /u/, shows that the English vowels are all significantly higher in the *F2* dimension than their Spanish counterparts, suggesting that the English vowels are all articulated with a fronted tongue position relative to the Spanish vowels.

This finding is consistent with other cross-language comparisons of acoustic vowel spaces (e.g., Disner, 1983), which find that the vowels of one language may differ in a systematic way from similar vowels of another language. This cross-linguistic difference has been accounted for by the notion of a language-specific base-of-articulation property, which is an important aspect of the description of the sound system of a language, and serves, in part, to differentiate systematically the general phonetic quality of two languages that may share certain phonemic categories. In other words, vowel categories that have the same phonological feature specifications and that occupy similar positions in the acoustic space across two different languages may have different precise phonetic realizations due to different bases-of-articulation of each language.

As a possible source of this language-specific base-of-articulation property, Honikman (1964) proposes that the base-of-articulation of a language is determined by the articulatory setting of the most frequent segments and segment combinations of the language. For example, she proposes that the base-of-articulation for British English is the cardinal alveolar tongue position, such as for /t,d,n,s,z/, since the alveolar place of articulation is the most frequent in the language. This explanation is consistent with the finding of the present study that the English vowels generally have higher *F2* values, and thus more fronted (that is, alveolarlike) tongue positions, than the corresponding Spanish vowels. However, since coronal segments occur with a high frequency in many (or perhaps most) languages, a careful study of the acoustic characteristics of a language whose most frequent segments are noncoronal is needed in order to verify this proposed source of the base-of-articulation of a language. For example, for languages such as Arabic and !Xóó, which have a preponderance of back segments, Honikman's proposal predicts a back base-of-articulation that should be

exhibited by relatively high *F1* values in the acoustic domain.

An implication for theories of vowel inventories of the observed *F2* difference between English and Spanish vowels is that, in order to assess the effect of universal factors on the acoustic realization of vowels, such a language-specific property must be taken into account. Accordingly, this finding contradicts the predictions of theories of vowel inventories that propose consistent locations in the acoustic space for similar phonemic vowel categories across different languages. For example, in assessing the conformity of acoustic data to the predictions of DT, the absolute location of vowel categories may not be an effective measure of dispersion; rather, factors such as the area covered by the vowels, or the relative arrangement of the vowels in the acoustic space, may provide more accurate measures of dispersion.

### B. Dispersion theory (DT)

The guiding principle behind DT is that vowels will tend to be maximally, or sufficiently, dispersed in the acoustic space in order to minimize the potential for perceptual confusion between separate vowel categories. Within this theory it usually been assumed that the boundaries of the available acoustic space are defined universally and that the distance between the separate vowel category locations is a measure of overall dispersion (e.g., Liljencrants and Lindblom, 1972; Lindblom, 1975, 1986). However, as demonstrated by the present data and other cross-linguistic acoustic studies (e.g., Lindau and Wood, 1977; Disner, 1983), the boundaries of the acoustic vowel space are more accurately defined on a language-specific basis.

With regard to the general principle of DT, the results of the present study provide some indication of a positive correlation between inventory size and area covered in the vowel space; however, this correlation is variable depending on the syllable context of the English vowels. In the closed syllable context the English /i/-/e/-/o/-/u/ quadrilateral covers more area than the corresponding Spanish or Greek quadrilaterals. However, in the open syllable context the English, Spanish, and Greek /i/-/e/-/o/-/u/ quadrilaterals all cover comparable areas. A possible source of the difference in area covered by the English vowels across syllable contexts could be the distribution of English vowels in closed versus open syllables. Both tense and lax vowels can occur in closed syllables, but only tense vowels can occur in open syllables (Ladefoged, 1993). That is, in the closed syllable, CVC context the English vowel space includes more categories than in the open syllable, CVCV context. In fact, in open syllables English has exactly the same number of categories as Spanish and Greek in the /i/-/e/-/o/-/u/ region of the acoustic



vowel space. Thus it could be that the /i/-/e/-/o/-/u/ areas vary both within and across languages according to the number of phonological categories that occupy this region of the vowel space. Alternatively, the difference in area covered by the English vowels in open and closed syllables could be a general syllable effect that operates independently of inventory size.

Another possible strategy for reducing the potential for intercategory confusion is to limit the degree of within-category spread. According to this strategy we would expect to find a positive correlation between inventory size and tightness of within-category clustering. However, the data of the present study did not show such a correlation; rather, the data showed no difference between the tightness of within-category clustering either within English (across syllable contexts), or across English and Spanish. It is possible that this result is attributable to the limited environment in which the vowels of these two languages were examined. In other words, it is possible that a difference in the degree of within-category clustering in English and Spanish would emerge in a study that systematically varied the segmental environment in which the target vowels appear.

An example of such a study is Manuel (1990) who investigated the effect of inventory size and structure on the degree of vowel-to-vowel coarticulation in three African languages, Ndebele, Shona, and Sotho. The results of that study show that the two languages with five vowels, Ndebele and Shona, have greater anticipatory coarticulation for the low vowel /a/ than does the language with seven vowels, Sotho.<sup>5</sup> However, for the midvowels these data show no difference in the degree of vowel-to-vowel coarticulation between the five- and seven-vowel systems. It is possible, however, that a more consistent difference would emerge in a similar comparison of languages such as English and Spanish, which have a larger difference in inventory size. Such a result would provide a strong indication that limiting the degree of within-category clustering is an attested means of ensuring sufficient dispersion.

### C. Quantal theory (QT)

In contrast to DT, which proposes that the locations of vowel categories in the acoustic space are determined by a principle of sufficient dispersion and are thus somewhat variable across languages, QT proposes that there are acoustically stable regions that operate universally. Thus according to QT, there are certain vowels, specifically the point vowels, whose locations in the acoustic space should be constant across all languages. However, the finding that the second formant frequency of the English high front vowel is significantly different from that of the Spanish high front vowel disconfirms the prediction of QT that the high front (that is, the low *F*<sub>1</sub> and high *F*<sub>2</sub>) corner of the acoustic space is a strictly quantal region that represents a stable phonemic category across languages. Rather, the present data indicate that, within this corner of the acoustic space, there is a certain degree of flexibility regarding the precise location of the /i/ category across different languages. Thus, in order to maintain the general principle of QT, at least as it pertains to the location of vowels in the acoustic space, we would have to

posit relatively stable regions as opposed to absolutely stable regions. Furthermore, in accordance with the results of a study by Pisoni (1980), the present data do not show any clear pattern of less variability for the point vowels as opposed to the nonpoint vowels. In other words, the present data do not support the claim that there are certain stable regions of the acoustic vowel space. Rather, the data show considerable variability across the two languages regarding the locations in the acoustic space of all of the shared vowel categories.

In conclusion, the results of this comparison of English and Spanish vowels suggest that the location in the acoustic vowel space of the vowel categories is determined, in part, by a language-specific, base-of-articulation property. In addition, the data suggest that syllable structure, rather than overall inventory size, determines the degree of expansion of the acoustic vowel space. Finally, the data show that the tightness of within-category clustering does not differ for the language with a crowded vowel space and the language with an uncrowded vowel space.

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<sup>1</sup>However, see Mendez (1982) for a contradicting result.

<sup>2</sup>A comparison of the formant values for both the English and Spanish vowels in this study with those from previous studies shows that the present measurements are similar to other data reported in the literature (e.g., Peterson and Barney, 1952; Delattre, 1969; Mendez, 1982).

<sup>3</sup>A comparison of the *F*<sub>1</sub> and *F*<sub>2</sub> measurements from the di- and monosyllabic English tokens shows that the locations of the target vowels in the acoustic space are affected by the syllabic structure of the word in which they appear. Specifically, the *F*<sub>2</sub> frequency in the CVC condition is higher than in the CVCV condition [*F*(1,10)=7.671, *p*=0.0076 by a two-factor ANOVA with CV pattern (CVC, CVCV) and vowel (the 11 English vowels) as factors]. Thus it appears that a syllable-final /t/ raises the *F*<sub>2</sub> frequency of the preceding vowel relative to the *F*<sub>2</sub> frequency in an open syllable. Nevertheless, the relevant comparison for this study is between the English and Spanish CVCV tokens.

<sup>4</sup>Two-factor ANOVAs with vowel (point versus nonpoint) and syllable type (CVC vs CVCV) as factors, and *F*<sub>1</sub> and *F*<sub>2</sub> as dependent variables, yield no significant main effects or interactions.

<sup>5</sup>Note that a recent acoustic and phonological study of Sotho by Khabanyane (1991) claims that Sotho has nine phonemic vowel categories.

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